

Ru⁺ ionisation limit 59366.40 cm⁻¹ = 7.36050 eV (168.445 nm)

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
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Transition	λ vac (nm)	λ air (nm)	(THz)	[J(upper)]	E (eV)	A (s^{-1})	F/2 π up (MHz)	branch (%)	Use
$^4F_3 - ^4F_6^o$	287.5832	287.4988	1042.4548	2.334	4.220e+07	11.37	59.1	first step of the two-colour two-photon resonance-ionizat...	
$^4F_3 - ^4F_5^o$	296.6032	296.5166	1010.7527	1.495	2.480e+07	10.33	38.2	high-resolution crossed-beam laser spectroscopy line	
$^4F_4 - ^4F_6^o$	297.7794	297.6925	1006.7603	1.086	8.220e+06	11.37	11.5	high-resolution crossed-beam laser spectroscopy line	
$^4F_5 - ^4F_6^oD_1^o$	298.9819	298.8947	1002.7110	1.723	2.500e+07	10.9	36.5	high-resolution crossed-beam laser spectroscopy line	
$^4F_5 - ^4F_5^o$	299.5841	299.4967	1000.6955	1.141	1.090e+07	10.27	16.9	high-resolution crossed-beam laser spectroscopy line	
$^4F_5 - ^4F_5^o$	300.7462	300.6586	996.8288	1.237	2.280e+07	9.474	38.3	high-resolution crossed-beam laser spectroscopy line	
$^4F_5 - ^4F_5^o$	302.1752	302.0733	992.1146	1.130	1.340e+07	10.23	20.6	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	304.1199	304.0314	985.7707	0.631	4.100e+06	9.942	10.2	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	304.3664	304.2478	985.0694	0.890	1.140e+07	6.372	19.1	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	304.9675	304.8788	983.0309	0.514	3.770e+06	6.831	8.8	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	310.1793	309.9999	970.6161	1.201	1.090e+07	10.9	15.9	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	316.0844	315.9929	948.4570	0.849	9.240e+06	8.246	17.8	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^oD_1^o$	318.6965	318.6043	940.6834	0.709	1.050e+07	8.466	19.7	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^oD_1^o$	319.0901	318.9979	939.5229	0.903	7.260e+06	8.246	14.4	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^oD_1^o$	319.7520	319.6605	937.5754	-	-	-	-	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	329.5061	329.4112	909.8236	0.696	3.050e+06	13.37	3.63	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^oG_2^o$	330.2545	330.1594	907.7620	0.721	2.660e+06	16.93	2.5	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^oG_2^o$	343.7721	343.6736	872.0675	4.007	7.280e+07	16.93	68.4	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^oG_2^o$	349.943	349.8942	856.5637	4.867	8.610e+07	13.72	99.9	reference line for collinear laser spectroscopy of Ru cha...	
$^4F_6 - ^4F_5^o$	359.0238	358.9213	835.0212	3.225	9.110e+07	15.45	93.8	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	359.4043	359.3017	834.1371	3.620	8.170e+07	15.6	83.3	high-resolution crossed-beam laser spectroscopy line	
$^4F_6 - ^4F_5^o$	359.7204	359.6178	833.4041	2.947	4.200e+07	19.24	68	high-resolution crossed-beam laser spectroscopy line	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)									
a ⁺ F _u	-229.288(74)	158.9(6)	F=5/2: +3508.9	F=7/2: +2657.47	F=9/2: +1582.77	F=11/2: +298.37	F=13/2: -1178.23	F=15/2: -2826.37				
a ⁺ F _v	-181.474(4)	101.8(5)	F=3/2: +2343.42	F=5/2: +1854.74	F=7/2: +1182.04	F=9/2: +436.77	F=11/2: -666.34	F=13/2: -1809.29				
a ⁺ F _w	-153.350(4)	59.32(5)	F=1/2: +1549.10	F=3/2: +1306.05	F=5/2: +906.91	F=7/2: +360.58	F=9/2: -320.50	F=11/2: -1120.30				
a ⁺ F _x	-92.4974(27)	31.869(23)	F=1/2: +669.79	F=3/2: +516.70	F=5/2: +269.52	F=7/2: -59.79	F=9/2: -454.52					
a ⁺ F _y	-116.93(20)	39.0(3)	F=3/2: -391.43	F=5/2: -150.81	F=7/2: +308.82							
c ⁺ G ₀	-97.10(10)	35.0(4)	F=7/2: +1838.30	F=9/2: +1294.08	F=11/2: +672.63	F=13/2: +0.17	F=15/2: -692.33	F=17/2: -1326.10				
c ⁺ G ₁	-128.8(5)	125.9(5)	F=3/2: +1671.48	F=5/2: +1312.59	F=7/2: +824.23	F=9/2: +220.49	F=11/2: -480.54	F=13/2: -1269.70				
c ⁺ G ₂	-74.50(20)	23.7(6)	F=5/2: +1220.03	F=7/2: +887.51	F=9/2: +488.38	F=11/2: +43.92	F=13/2: -419.82	F=15/2: -872.10				
c ⁺ G ₃	-87.3(3)	16.9(5)	F=3/2: +1099.25	F=5/2: +876.20	F=7/2: +565.76	F=9/2: +169.77	F=11/2: -309.42	F=13/2: -668.92				
c ⁺ G ₄	-123.4(3)	178.9(27)	F=1/2: +1341.34	F=3/2: +1107.94	F=5/2: +736.82	F=7/2: +254.83	F=9/2: -300.47	F=11/2: -880.77				
c ⁺ G ₅	-249.1(5)	162.8(28)	F=1/2: +1857.66	F=3/2: +1410.75	F=5/2: +706.60	F=7/2: -193.74	F=9/2: -1204.80					
d ⁺ D ₀	-149.30(10)	-149.7(28)	F=1/2: +1403.18	F=3/2: +1219.65	F=5/2: +898.79	F=7/2: +418.16	F=9/2: -253.69	F=11/2: -1157.17				
d ⁺ D ₁	-175.20(20)	-11(4)	F=3/2: +2184.75	F=5/2: +1749.90	F=7/2: +1131.91	F=9/2: +353.57	F=11/2: -610.66	F=13/2: -1754.67				
d ⁺ D ₂	-117.80(10)	-47.0(6)	F=1/2: +791.70	F=3/2: +636.15	F=5/2: +365.15	F=7/2: -38.92	F=9/2: -600.75					
d ⁺ D ₃	-229.5(3)	18(7)	F=3/2: +2305.56	F=5/2: +1956.56	F=7/2: +1376.65	F=9/2: +568.47	F=11/2: -464.28	F=13/2: -1716.85				
d ⁺ D ₄	-109.70(10)	39.6(9)	F=3/2: +411.67	F=5/2: +78.02	F=7/2: -264.35							
e ⁺ F ₀	-221.4(4)	491(17)	F=5/2: +3533.77	F=7/2: +2609.93	F=9/2: +1481.06	F=11/2: +191.35	F=13/2: -1205.20	F=15/2: -2644.75				
e ⁺ F ₁	-162.3(4)	-50(6)	F=1/2: +1100.75	F=3/2: +880.02	F=5/2: +499.52	F=7/2: -59.69	F=9/2: -824.12					
e ⁺ F ₂	-147.40(20)	271(4)	F=3/2: +1975.48	F=5/2: +1527.19	F=7/2: +930.05	F=9/2: +214.53	F=11/2: -580.22	F=13/2: -1406.30				
e ⁺ F ₃	-123.9(4)	136(8)	F=1/2: +1320.72	F=3/2: +1098.10	F=5/2: +740.68	F=7/2: +268.89	F=9/2: -288.66	F=11/2: -895.20				
e ⁺ F ₄	-82.4(4)	96(10)	F=1/2: +643.72	F=3/2: +477.10	F=5/2: +223.30	F=7/2: -81.83	F=9/2: -388.10					

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^3F_4 - y^3F_4$	296.603	102-104	79.5(16)
$a^3F_4 - y^3F_4$	296.603	101-104	103(4)
$a^3F_4 - y^3F_4$	296.603	100-104	168.2(28)
$a^3F_4 - y^3F_4$	296.603	99-104	193(5)
$a^3F_4 - y^3F_4$	296.603	98-104	264.6(25)
$a^3F_4 - y^3F_4$	296.603	96-104	341(4)
$a^3F_4 - y^3F_4$	297.779	102-104	-11(8)
$a^3F_4 - y^3F_4$	297.779	101-104	-37(8)
$a^3F_4 - y^3F_4$	297.779	100-104	-5(15)
$a^3F_4 - y^3F_4$	297.779	99-104	7(8)
$a^3F_4 - y^3F_4$	297.779	98-104	40(16)
$a^3F_4 - y^3D_4$	297.779	96-104	27(2)
$a^3F_4 - y^3D_4$	298.982	102-104	137.6(10)
$a^3F_4 - y^3D_4$	298.982	101-104	213.6(51)
$a^3F_4 - y^3D_4$	298.982	100-104	277.2(14)
$a^3F_4 - y^3D_4$	298.982	99-104	356.0(17)
$a^3F_4 - y^3D_4$	298.982	98-104	426.4(14)
$a^3F_4 - y^3D_4$	298.982	96-104	579.6(19)
$a^3F_4 - y^3F_4$	299.584	102-104	46(5)
$a^3F_4 - y^3F_4$	299.584	101-104	44(3)
$a^3F_4 - y^3F_4$	299.584	100-104	110(4)
$a^3F_4 - y^3F_4$	299.584	99-104	111(5)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$
